

FA7 - PART 2

Applying Normal Distribution on Campus

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Time people spent in Freedom Park



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Fa7 part 2

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⋮

B 2 — 15:19 - 15:40

✓

C 1 — 15:20 - 15:21

✓

A 3 — 15:20 - 16:20

✓

1 — 15:24

✓

1 — 15:34 - 16:20

✓

C 2 — 15:26 - 16:07

✓

C 1 — 15:28 - 15:47

✓

A 1 — 15:29 - 15:34

✓

C 1 — 15:33 - 15:34

✓

B 1 — 15:34 - 16:09

✓

Observation period:
Oct 01, 2026
15:19 - 16:56

	A	B	C	D
1	PEOPLE	START	END	MINUTES_SPENT
2	2	15:19	15:40	21
3	1	15:20	15:21	1
4	3	15:20	16:20	60
5	1	15:20	15:24	4
6	1	15:34	16:20	46
7	2	15:26	16:07	41
8	1	15:28	15:47	19
9	1	15:29	15:34	5
10	1	15:33	15:34	1
11	1	15:34	16:09	35
12	1	15:35	15:36	1
13	1	15:36	15:42	6
14	3	15:38	16:04	26
15	2	15:40	15:59	19
16	2	15:41	15:50	9
17	1	15:45	15:54	9
18	1	15:48	16:13	25
19	1	15:49	16:44	55
20	1	15:53	16:05	12

*** A, B, and C are areas divided to help me keep track of people easily.
*** Not all data are shown.

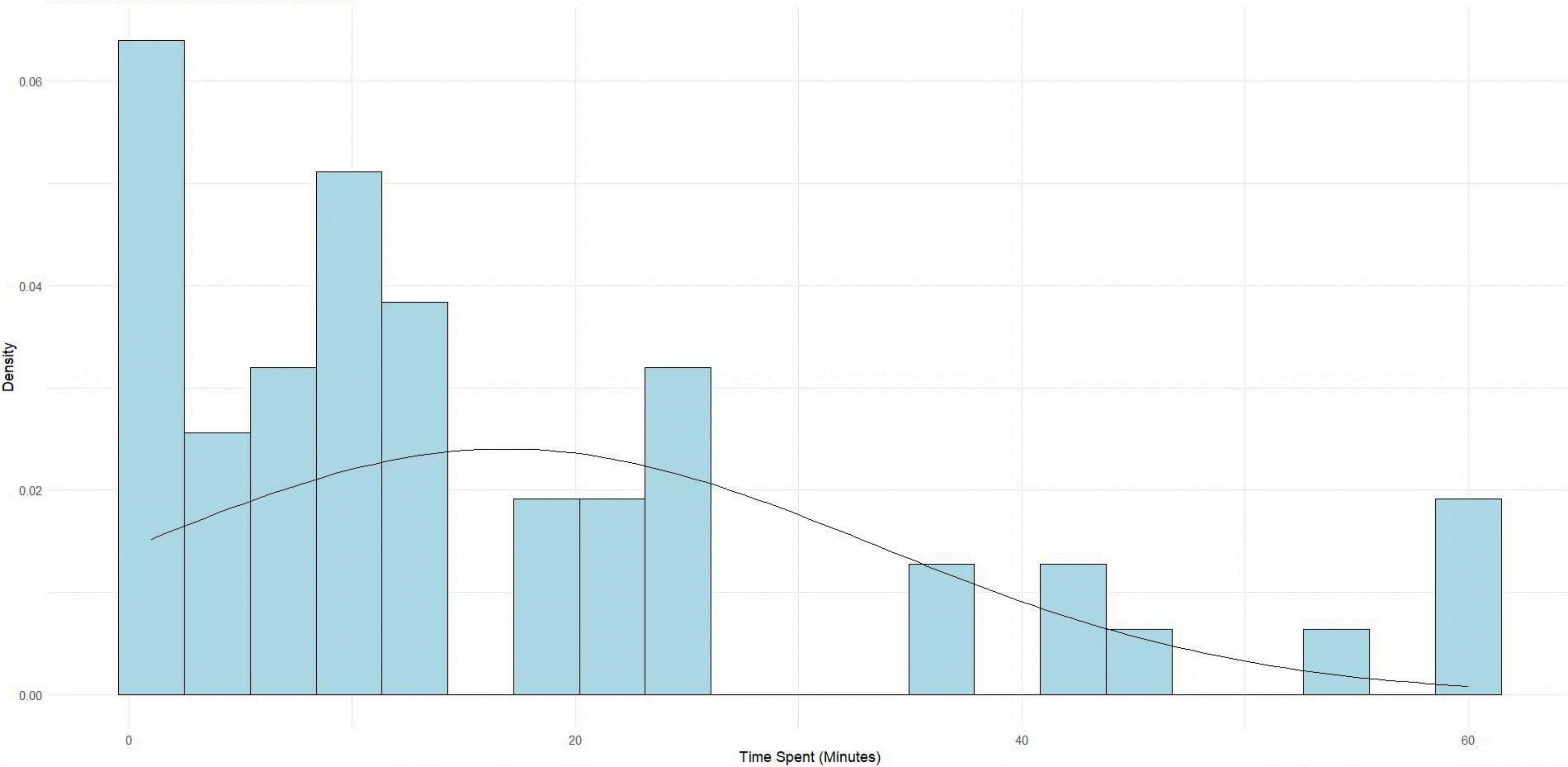
Statistics

```
162 Number of people: 53
163
164 Mean time spent: 16.8868 minutes
165
166 Standard deviation: 16.6001 minutes
167
168 Median: 11 minutes
169
170 Minimum: 1 minute
171
172 Maximum: 60 minutes
```

```
> print(frequency_table)
```

	MINUTES_SPENT	FREQUENCY	PERCENTAGE
1	1	6	11.320755
2	2	4	7.547170
3	3	1	1.886792
4	4	1	1.886792
5	5	2	3.773585
6	6	4	7.547170
7	8	1	1.886792
8	9	4	7.547170
9	10	2	3.773585
10	11	2	3.773585
11	12	6	11.320755
12	19	3	5.660377
13	21	3	5.660377
14	25	2	3.773585
15	26	3	5.660377
16	35	1	1.886792
17	36	1	1.886792
18	41	2	3.773585
19	46	1	1.886792
20	55	1	1.886792
21	60	3	5.660377

Time People Spent in Freedom Park
Observed Data with Normal Distribution Curve



```
> # Percentage of Data Within 1, 2, and 3 Standard Deviation
```

```
> # Within 1 sd
```

```
> strd1 <- sum(
```

```
+   duration_data >= mn - sd &
```

```
+   duration_data <= mn + sd
```

```
+ ) / totalp * 100
```

```
> sd1 <- round(strd1, 2)
```

```
> sd1
```

```
[1] 83.02
```

```
>
```

```
> # Within 2 sd
```

```
> strd2 <- sum(
```

```
+   duration_data >= mn - 2 * sd &
```

```
+   duration_data <= mn + 2 * sd
```

```
+ ) / totalp * 100
```

```
> sd2 <- round(strd2, 2)
```

```
> sd2
```

```
[1] 92.45
```

```
> # Within 3 sd
```

```
> strd3 <- sum(
```

```
+   duration_data >= mn - 3 * sd &
```

```
+   duration_data <= mn + 3 * sd
```

```
+ ) / totalp * 100
```

```
> sd3 <- round(strd3, 2)
```

```
> sd3
```

```
[1] 100
```

```
228 # Theoretical Normal Distribution:
```

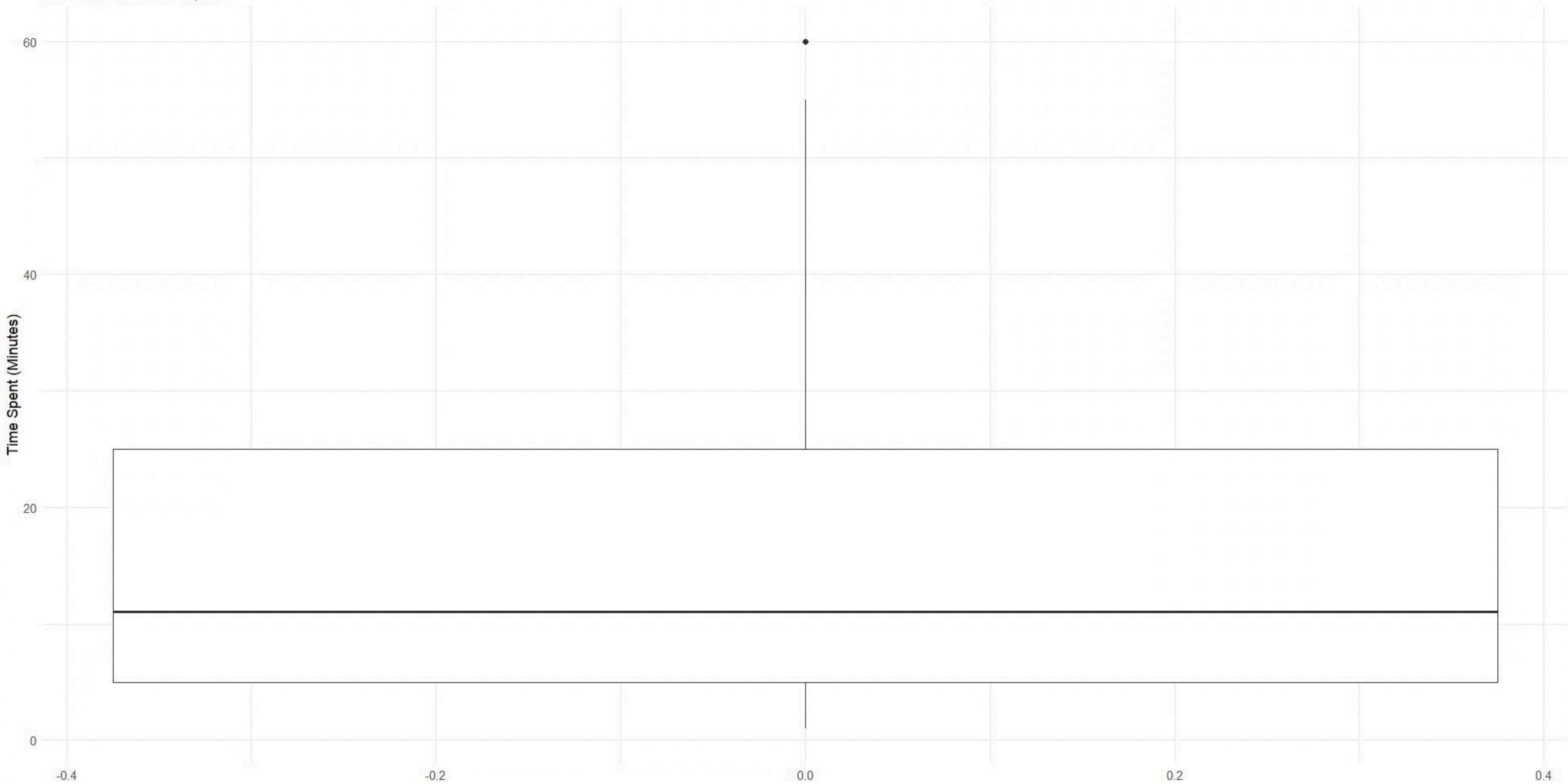
```
229 # Within  $1\sigma$  = 68.27%
```

```
230 # Within  $2\sigma$  = 95.45%
```

```
231 # Within  $3\sigma$  = 99.73%
```

NOT PERFECTLY normally distributed

Box Plot of Time Spent



```

> # Outlier
> # Q1
> q1 <- quantile(duration_data, 0.25)
> q1
25%
  5
>
> # Q3
> q3 <- quantile(duration_data, 0.75)
> q3
75%
 25
>
> # IQR
> iqr <- q3 - q1
> iqr
75%
 20

```

```

> # Lower Bound
> low <- q1 - 1.5 * iqr
> low
25%
-25
>
> # Upper Bound
> up <- q3 + 1.5 * iqr
> up
75%
 55
>
> # Number of Outliers
> outliers <- duration_data[
+   duration_data < low |
+   duration_data > up
+ ]
> ol <- length(outliers)
> ol
[1] 3
>
> if (length(outliers) > 0) {
+   cat("\nOutlier values:\n")
+   print(outliers)
+ } else {
+   cat("\nNo outliers detected using the 1.5 x IQR rule.\n")
+ }

```

```

Outlier values:
[1] 60 60 60

```

```

> # Is distribution symmetric?
> mn
[1] 16.8868
> med
[1] 11
>
> if (mn > med) {
+   cat("\nThe distribution is likely right-skewed.")
+ } else if (mn < med) {
+   cat("\nThe distribution is likely left-skewed.")
+ } else {
+   cat("\nThe mean and median are equal, suggesting symmetry.")
+ }

The distribution is likely right-skewed.

```

Real-Life Implications and Recommendations

- The results is for understanding how people used the area
- Campus planners should maintain and improve this space
- Longer observation periods can improve can be more accurate in studying this topic

